

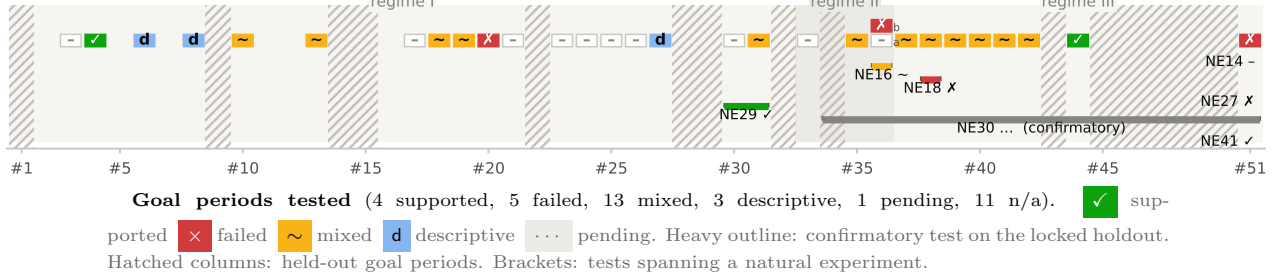
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Memory and the agent’s repo carry no measurable load-bearing information; a forced context wipe costs 39% of committed work for ten calls, whatever the agent saves.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

The data is natural scrambles in non-holdout periods, per goal period, against a placebo-day null. A synthetic check on the real village skeleton calibrates the null (power 17–41%). **Memory:** 12 losses of 50–80% of an agent’s memory and 20 newcomers with empty memories show no day-scale cost on the viability V^* . In round 1b memory loss gives +0.29 SD $[-0.05, +0.63]$, and newcomers give +0.20 $[-0.13, +0.54]$: they progress *more*. The rewrite control is null. **Chat cuts:** only 3 usable events exist. **Context:** consolidations wipe the context window (forced every 41 turns) but keep memory. The pre-registered forced-against-voluntary contrast has the wrong sign (task-phase confound). Post hoc, a forced wipe lowers output for about ten calls, whatever the agent saves to memory. On the round-1b work ledger the cost is –39% of work commits for ten calls (7/9 periods; about 10% of output). D2.6 finds no homeostatic V in the pre-registered pool. The holdout has not run.

PHYSICAL MODEL

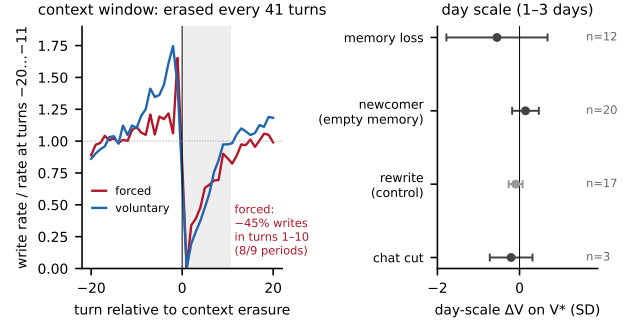
Model 04, Kolchinsky–Wolpert semantic information. The system X is one agent: its memory M (day-scale store) and its context window C (session-scale store, filled by chat and events). Y is the rest of the village. A scramble erases M or C , or cuts $Y \rightarrow X$; $\hat{p}$ is the scrambled joint. V is viability per agent-day (5 candidates) or per call (work). u is its same-day residual, and δ is the fraction removed. S is the semantic information and η its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} | u_{\text{pre}}])$$



Observables. Round 1. Left: the write rate around consolidations, relative to turns $-20 \dots -11$ (regime III, pooled). A forced consolidation (41-turn cap, timing independent of task state) wipes the context. Output falls ~45% for ten turns, then recovers (post hoc). Voluntary consolidations dip alike after a write burst. Right: the day-scale ΔV on V^* (meta over goal periods, 95% CI). Memory loss, newcomers, the rewrite control and chat cuts are all consistent with zero.

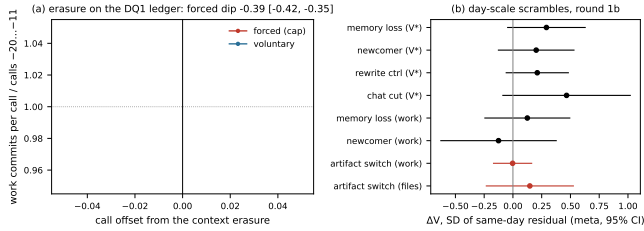
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The working context keeps an agent productive. The agent rebuilds a long-term memory file from its environment within a day. That confirms what operators of LLM agents already suspect. The usable number is the cost of a context reset: –39% of committed work for about ten calls, about a tenth of output at a 41-turn cadence. To recover it, raise the cap or carry a task-state summary across the reset. Today’s memory write does not recover the loss. That is scaffold engineering, not physics, and post hoc until the holdout check (C3) runs. Kolchinsky–Wolpert adds vocabulary (stored against observed information), but here it measures neither S nor η , nor a viability function.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (im- proved data)	Verdict
D2.6 choose V^*	I: $V_{\text{rel}} \rightarrow V_{\text{eng}}$; III: $V_{\text{eng}} \rightarrow V_{\text{rel}}$ (real failures); V_{files} homeostatic (new)	inconcl.
P1 memory loss ≤ -0.25	$-0.55 [-1.78, 0.68] \rightarrow +0.29 [-0.05, 0.63]$	$\rightarrow \times$
P3 rewrite control null	$-0.10 \rightarrow +0.21$ ($z -1.1 \rightarrow +1.5$)	$\checkmark$
P4 newcomers pay	$+0.15 \rightarrow +0.20$; progress $+0.31 [0.06, 0.56]$	$\times$
P5 forced $<$ voluntary	writes $z + 4.9 \rightarrow \text{work } z + 2.3$	$\times$
P5' forced-wipe dip	writes -0.44 (8/9) $\rightarrow \text{work } -0.39 [-0.42, -0.35]$ (7/9)	$\checkmark$ (NE41)
P6 chat cut harmless	$-0.72 \rightarrow -0.59$ (3 events)	$\times$
P11 artifact switch costs (new)	work $-0.00 [-0.17, 0.17]$; $\kappa_{\text{AS}} \approx 0 < \kappa_{\text{ML}}$	$\times$
NE41 memory dose (native)	$\rho = +0.01$; high-low tercile $-0.05 [-0.10, 0.01]$	$\checkmark$
NE29 lineage retired incumbents' work (native)	$z = +0.13$	$\checkmark$



Second figure. Round 1b. (a) Work commits per call around a context erasure on the DQ1 ledger, relative to calls $-20 \dots -11$. A forced wipe (41-turn cap) cuts committed work by about 39% for ten calls. Voluntary consolidations dip more after a work burst (task phase). (b) Day-scale scrambles on V^* and on work output (meta over goal periods, 95% CI): memory loss, newcomers, the rewrite control, chat cuts and artifact switches. Only the chat cut (3 events) departs from zero.

GOAL PERIODS IN DETAIL

NE30	...	pending (confirmatory; not run) (holdout)
G04	$\checkmark$	1b: supported ($V^* V_{\text{eng}}$; MN -0.50 ($z -0.7$); 2026-10-0...
G06	d	1b: descriptive ($V^* V_{\text{eng}}$; MR $+0.40$ ($z +0.5$); 2026-10...
G08	d	1b: descriptive ($V^* V_{\text{eng}}$; MR -0.69 ($z -1.0$); 2026-10...
G10	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.49$ ($z +1.5$); 2026-10-04, Γ ...
G13	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; ML $+0.40$ ($z +1.2$); 2026-10-04, Γ ...
G18	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.58$ ($z +1.2$); 2026-10-04, Γ ...
G19	$\sim$	1b: mixed ($V^* V_{\text{eng}}$; MN $+0.96$ ($z +1.2$); 2026-10-04, Γ ...
G20	$\times$	1b: failed ($V^* V_{\text{eng}}$; MN $+0.56$ ($z +1.1$); 2026-10-04, ...
G27	d	1b: descriptive ($V^* V_{\text{eng}}$; MR $+0.10$ ($z +0.2$); 2026-10...

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The day-scale tests stay underpowered (synthetic power 17–41%). V^* moves in both regimes with the corrected reliability measure, so the period verdicts are replications under a changed yardstick. 118 of 172 artifact switches fall on a goal's first day, where the within-period estimator cannot reach them. The agents choose the other switches, as they choose voluntary consolidations. The erasure dip is post hoc in origin. NE41 replicates it with dated predictions on independent data, but the holdout check (C3) has not run. V_{files} passes D2.6 on 5 shocks and is outside the pre-registered pool.

ROUND 2

Re-freeze `confirm_ne30.py` on work commits and run C3. Pre-register V_{files} on new shocks.

- **Essence.** Where does the swarm keep its knowledge? In artifacts and history, not in agents.
- **H15-R1** Stigmergic memory: after a memory loss, agents re-acquire state from artifacts (file reads, history search) within a few turns; richer artifact trails mean faster recovery (E3).
- **H15-R2** Semantic information is concentrated in the few items read first after a reset (the goal and the last messages).
- **H15-R3** Memory notes are performative: their content rarely reappears in later actions.

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